

GLMs for Applied Research

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Session 8: Introduction to mixed effects models

Topics we will cover in this session

- From single-level GLMs to clustered data
- Fixed vs. random effects
- Random intercepts vs. random slopes
- The mixed-effects GLM equation
- Fitting in R, testing the diagnostics, interpreting the output
- Extending to counts, pitfalls & takeaways

From single-level GLMs to clustered data

The GLMs framework

- A link function $g()$ connects a linear predictor to the mean of an outcome from the exponential family

$$g(E[Y]) = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k$$

- Logistic, multinomial, ordinal, and Poisson regression are all GLMs — they differ only in family and link
- Every observation is assumed independent, given the predictors

But real data are often clustered

Grouped observations: animals within herds

15 cattle herds, disease incidence tracked over 4 time periods each (the cbpp study)

Repeated measures: the same subject over time

18 subjects, reaction time measured on 10 consecutive days of sleep deprivation (the sleepstudy data)

Clustered data violate the independence assumption

When observations are grouped — animals within herds, patients within hospitals, repeated measures within subjects — observations in the same group tend to be more similar to each other than to observations in other groups. Ignoring this is called pseudoreplication.

What goes wrong

A standard GLM treats every row as an independent unit of information. Clustered data have fewer truly independent pieces of information than rows, so SEs are too small.

The consequence

Overly narrow confidence intervals and inflated test statistics — you'll find “significant” effects more often than the data actually justify.

The fix

Add random effects that explicitly model the group structure, so the model “knows” which rows are correlated and adjusts uncertainty accordingly.

Example

The cbpp dataset: contagious bovine pleuropneumonia incidence in 15 African cattle herds, tracked over 4 time periods — 56 herd-period observations, 842 animals in total.

Fixed vs. random effects

1 Fixed effects

- Parameters you want to estimate directly, one value each
- The levels observed are the levels you care about (e.g., treatment period 1-4)
- Same interpretation as an ordinary GLM coefficient
- Example: the effect of disease-control period on infection odds

2 Random effects

- Group-level deviations you want to account for, not estimate individually
- Observed groups are treated as a sample from a larger population of groups
- Assumed to follow a distribution, usually $\text{Normal}(0, \sigma^2)$
- Example: herd-to-herd variation in baseline disease risk

Random intercepts vs. random slopes

Both let each group deviate from the population pattern — but in different ways:

Random intercept only



Each group starts at a different baseline — but responds to the predictor with the SAME slope (parallel lines).

Random intercept + random slope



Each group also differs in HOW STRONGLY it responds — slopes fan out rather than stay parallel.

Real example (sleepstudy): reaction time rises 10.5 ms/day of sleep deprivation on average — but the between-subject SD in that slope is 5.9 ms/day, so some people are hit far harder than others.

The mixed effect model

A mixed-effects GLM keeps the familiar link function, but the linear predictor now has a fixed part shared by everyone and a random part that varies by group j :

$$g(E[Y_{ij} \mid b_j]) = X_{ij}\beta + Z_{ij} b_j \quad \text{where } b_j \sim N(0, D)$$

i indexes observations, j indexes groups (e.g., herds); b_j is group j 's random effect

$g()$: the link function

Same choice as an ordinary GLM — logit for binomial outcomes, log for Poisson counts, and so on.

$X\beta$: the fixed part

Population-average effects, identical for every group — estimated exactly as in a standard GLM.

Zb_j : the random part

Group-specific deviations from that population average, assumed drawn from a shared Normal distribution.

Fitting a binomial mixed model in R

cbpp: disease incidence out of herd size, by time period, with a random intercept for herd

```
library(lme4)
data(cbpp) # 15 herds x 4 periods, 56 rows, 842 animals
gm1 <- glmer(cbind(incidence, size - incidence) ~ period + (1 | herd),
             data = cbpp, family = binomial)
summary(gm1)      # fixed effects + random-effect variance
exp(fixef(gm1))   # odds ratios
```

cbind(incidence, size - incidence)

The standard “successes, failures” format for binomial data recorded as counts out of a total.

(1 | herd)

A random-intercept term: each herd gets its own baseline log-odds, drawn from a shared Normal distribution.

Fit by (approximate) ML

glmer() always maximizes an approximation to the marginal likelihood — more on this shortly.

Fixed effects, as Odds ratios

Real fitted results from gm1 — disease-incidence odds by period, relative to period 1:

Term	Estimate (log-odds)	Odds ratio	p-value
Intercept (period 1)	-1.398	0.247	<0.001
period 2	-0.992	0.371	0.001
period 3	-1.128	0.324	<0.001
period 4	-1.580	0.206	<0.001

Reading tip

Fixed effects are interpreted exactly like an ordinary logistic regression: the odds of CBPP incidence in period 4 are about 21% of the odds in period 1 (OR = 0.206), holding herd-level baseline risk fixed. All three period effects are significant — incidence odds decline steadily, consistent with disease-control measures taking hold over time.

Random effects and the ICC

Herd variance = 0.412 (SD = 0.642, on the logit scale)

$$\text{ICC} = \sigma^2_{\text{herd}} / (\sigma^2_{\text{herd}} + \pi^2/3) = 0.412 / (0.412 + 3.29) \approx 0.111$$

- The residual variance for a logit-linked binomial outcome is fixed at $\pi^2/3 \approx 3.29$, which is what makes this ICC formula work for logistic mixed models
- $\text{ICC} \approx 0.11$ means about 11% of the total latent variation in CBPP susceptibility is attributed to differences BETWEEN herds — modest, but not negligible
- A plain logistic regression (ignoring herd) would treat all 56 observations as fully independent, understating uncertainty in the period effects
- Report the random-effect variance (or ICC) alongside fixed-effect estimates — it's part of the finding, not just a nuisance parameter

Why glmer() Needs Approximation

Random effects are unobserved, so fitting the model means integrating them out of the likelihood — and for non-Gaussian outcomes, that integral has no closed form.

glmer() approximates it via the Laplace approximation (default), or Adaptive Gauss-Hermite Quadrature ($nAGQ > 1$) for more accuracy with simple random-effect structures.

REML vs. ML — a Gaussian-only Choice

- For linear mixed models (lmer()), REML corrects the downward bias ML gives variance components, and is the default.
- REML's derivation doesn't generalize cleanly to non-Gaussian outcomes, so glmer() always fits by (approximate) ML — there is no REML option for a GLMM.
- Use anova()/AIC to compare glmer() models fitted by ML; that comparison is valid, unlike comparing REML-fitted lmer() models with different fixed effects.

Testing whether you need the random effect

```
gm0 <- glm(cbind(incidence, size - incidence) ~ period, data = cbpp, family = binomial)
anova(gm1, gm0)  # likelihood ratio test: does adding (1|herd) help?
```

Result

AIC without herd random effect: 206.1
AIC with herd random effect: 194.1
→ a clear improvement from modeling herd-to-herd variation.

The boundary caveat

Testing $\sigma^2 = 0$ sits on the edge of the parameter space (variance can't go negative), so the usual chi-square reference is anti-conservative. Common fixes: halve the naive p-value, or use a simulation-based test like `RLRsim::exactRLRT()`.

If the study design implies clustering — repeated measures, known groups — include the random effect regardless of a marginal significance test. The random effect reflects your sampling design, not just a fit statistic.

Convergence warnings and singular fits

Convergence Warnings

- More common with complex random-effect structures or few groups per factor
- Try: center/scale continuous predictors, increase iterations (control), or compare optimizers with `lme4::allFit()`
- A warning doesn't always mean the fit is unusable — but it always deserves a second look

“Singular Fit” Warnings

- A random-effect variance is estimated at (or very near) zero, or a correlation at ± 1
- Sign that the random-effects structure is too rich for the data — not enough groups, or not enough variation to estimate it
- Fix by simplifying: drop random slopes first, then correlations between random effects, before dropping fixed effects

Residuals checks

Ordinary Pearson/deviance residuals are hard to interpret for GLMMs. DHARMA instead simulates new data from the fitted model many times and compares each observation to its simulated distribution — giving residuals that should be uniform if the model fits well.

```
library(DHARMA)
sim_res <- simulateResiduals(gm1)
plot(sim_res)          # QQ plot + residual-vs-predicted
testDispersion(sim_res) # over/underdispersion test
testZeroInflation(sim_res) # excess-zeros test (for count models)
```

What to look for

- A curved or S-shaped QQ plot: the chosen distribution family may be wrong
- Patterns in residuals vs. fitted values: a missing predictor, or the wrong link function
- A significant dispersion test: consider a negative-binomial or observation-level random effect for count data

Poisson & Negative Binomial mixed models

The same fixed + random structure extends directly to count outcomes — swap the link and family:

```
# Poisson mixed model (log link)
glmer(ticks ~ height + (1 | brood), data = grouseticks, family = poisson)

# Negative binomial, for overdispersed counts
glmer.nb(ticks ~ height + (1 | brood), data = grouseticks)
```

Real example: grouseticks

Number of ticks found on red grouse chicks, with brood and location random effects — a classic ecological field dataset bundled with lme4.

Overdispersion is common

Count data often show more variance than a Poisson implies. Check with `DHARMA::testDispersion()` before trusting Poisson SEs.

glmer.nb() or glmmTMB

Negative-binomial GLMMs handle overdispersion directly; glmmTMB offers additional families and zero-inflation options.

Common pitfalls to avoid

1 Too few groups

Estimating a variance component reliably usually needs at least 5-10 groups — fewer than that, consider treating the grouping variable as fixed instead.

2 Overly rich random effects

“Keep it maximal” (Barr et al., 2013) is good in principle, but an over-parameterized structure the data can't support leads to singular fits. Simplify when you hit one.

3 Unscaled predictors

Continuous predictors on very different scales are a common cause of convergence warnings — center and scale them before fitting.

4 Reporting only p-values

Always report the random-effect variance (or ICC) alongside fixed effects — it's part of the scientific story, not a nuisance parameter.

Recap

- Mixed-effects GLMs add group-level random effects to the familiar GLM framework, for clustered or repeated-measures data
- Fixed effects are population-average parameters, interpreted as usual; random effects capture group-level variability, summarized by a variance and the ICC
- Fit with `lme4::glmer()` (ML via Laplace/AGQ — there's no REML option for GLMMs); test random effects cautiously given the boundary problem
- Always check convergence, singular fits, and simulation-based residuals (DHARMa) before trusting the model

➤ **Next session:** Hands-on Mixed-effects GLMs